CS 6846: Quantum Computing

Homework 2

Name: Swathi Shree Narashiman *Due: 23rd September, 2024*

Collaborations:

1. Nitin G - EE22B041
2. Shri Prathaa - EE22B144

Problem 1: Superdense Coding (5 points)

Alice and Bob prepare an EPR pair, that is, two qubits in the state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. They each take one qubit home. Suddenly, Alice decides she wishes to convey one of 4 messages to Bob; in other words, she wants to convey a classical string $uv \in \{0,1\}^2$ to Bob. Alice does the following in the privacy of her own home: First, if $u = 1$, she applies a NOT gate to her qubit (else if $u = 0$ she does nothing here). Next, if $v = 1$, she applies a Z-gate to her qubit (else if $v = 0$, she does nothing here). Finally, she walks to Bob's house and silently hands him her qubit. Show that by measuring in an appropriate basis, Bob can exactly determine Alice's message $uv \in \{0,1\}^2$.

Solution:

Not get belongs to the following transformation:

Input	**Output**
$\|0\rangle$	$\|1\rangle$
$\|1\rangle$	$\|0\rangle$

Z gate belongs to the following transformation:

Input	**Output**
$\|0\rangle$	$\|0\rangle$
$\|1\rangle$	-$\|1\rangle$

Based on the constraints described in the problem we can formulate the following table to describe the 4 input cases:

Let's start with the EPR Pair, $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and let $|\phi\rangle$ denote the output state.

{u, v}	$\|\phi\rangle$
{0, 0}	$\|\phi_1\rangle = \frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$
{0, 1}	$\|\phi_2\rangle = \frac{1}{\sqrt{2}}(\|00\rangle - \|11\rangle)$
{1, 0}	$\|\phi_3\rangle = \frac{1}{\sqrt{2}}(\|01\rangle + \|10\rangle)$
{1, 1}	$\|\phi_4\rangle = \frac{1}{\sqrt{2}}(\|01\rangle - \|10\rangle)$

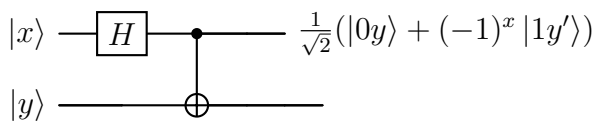
It's clear that each of these states is a bell state, and the 4 bell states make an orthonormal set. Hence, they form a basis for 2-bit qubit space. Therefore, if Bob chooses the 4 EPR pairs as the basis states, then he can precisely determine Alice's message.

The EPR Pairs can be expressed as:
$[\frac{1}{\sqrt{2}}, 0, 0, \frac{1}{\sqrt{2}}]^T, [\frac{1}{\sqrt{2}}, 0, 0, -\frac{1}{\sqrt{2}}]^T, [0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0]^T, [0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0]^T$

Clearly all the vectors are unit norm. The inner product of any two vectors are zero. (Therefore they are orthonormal).

Let output be $|\phi\rangle$, then
$\langle\phi_1|\phi\rangle = 1 =====> \{u, v\} = \{0, 0\}$
$\langle\phi_2|\phi\rangle = 1 =====> \{u, v\} = \{0, 1\}$
$\langle\phi_3|\phi\rangle = 1 =====> \{u, v\} = \{1, 0\}$
$\langle\phi_4|\phi\rangle = 1 =====> \{u, v\} = \{1, 1\}$

Also any bell state is prepared from a classical state $|xy\rangle$ as follows:



Hence, to get unique bits x and y, we need to perform the reverse operation, that is applying a CNOT followed by a Hadamard on the first bit. This gives us the computational basis of a 2bit qubit system. This is equivalent to measuring in the bell state basis.

Problem 2: Deferred Measurements (3 points)

For $0 \leq p \leq 1$, let COIN_p denote a gate that has no input and one output, with the output being a random bit which is 1 with probability p and 0 with probability $1 - p$. The standard way to augment the basic circuit model with randomness is to allow the use of $\text{COIN}_{1/2}$ gates.

In our definition of quantum circuits, we allowed quantum gates plus measurement at the very end. We saw in class that CCNOT can simulate the AND, OR, and NOT gates. To simulate $\text{COIN}_{1/2}$ gates, we suggested passing a $|0\rangle$ qubit through a Hadamard gate and then measuring it.

However, if we want to use this random bit within our circuit, we need to augment the quantum circuit model by allowing "intermediate measurements" (i.e., measuring some qubits prior to the end of the computation). While this is acceptable both theoretically and physically, it makes the model somewhat more complicated.

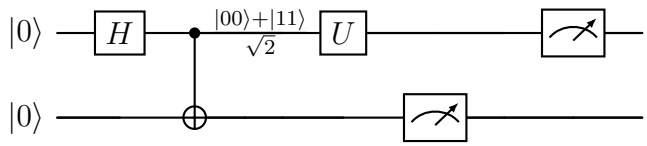
Luckily, we can show that any computation done by a quantum circuit using intermediate measurements can be equivalently and nearly as efficiently done by a quantum circuit that only has a single measurement at the end. In this problem, you won't quite prove this in full, but you'll get the essential idea. Precisely, suppose C is a randomized circuit with n input bits, a ancilla bits, r $\text{COIN}_{1/2}$ gates, s CCNOT gates, and m output bits (possibly including garbage).

Describe a straightforward transformation to a quantum circuit C' with n input bits, $a + 2r$ ancilla bits, $s + 2r$ CCNOT/CNOT/Hadamard gates, and $m + r$ output bits, such that when the output bits are measured at the end of $C'(x)$, the probability distribution on the first m of them is exactly the same as the probability distribution on the output bits of $C(x)$. (Prove that your circuit indeed produces random bits.)

Solution:

We can replace the each COIN gate, using an IPR pair and measure the state of the second bit .

For example consider the circuit,



Since the states are entangled, we can perform computations on the first bit and perform measurement at the end on the second bit.

1. Ouput of a $COIN_{1/2}$ gate is say $|\psi\rangle$. Let it undergo some unitary transformations in the circuit (say U).

   At the end when we measure in computation basis,

$$P(0) = |\langle 0|U\psi\rangle|^2 = |\langle U^\dagger 0|\psi\rangle|^2 = P(\psi = 0)|\langle U^\dagger 0|0\rangle|^2 + P(\psi = 1)|\langle U^\dagger 0|1\rangle|^2$$
$$P(0) = \frac{1}{2}|\langle U^\dagger 0|0\rangle|^2 + \frac{1}{2}|\langle U^\dagger 0|1\rangle|^2$$
$$P(1) = |\langle 1|U\psi\rangle|^2 = |\langle U^\dagger 1|\psi\rangle|^2 = \frac{1}{2}|\langle U^\dagger 1|0\rangle|^2 + \frac{1}{2}|\langle U^\dagger 1|1\rangle|^2$$

2. Consider the entangled state now,

$$|\psi'\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

   Let us apply the transformation on the first bit now,

$$|\psi''\rangle = \frac{1}{\sqrt{2}}(|U0\rangle\,|0\rangle + |U1\rangle\,|1\rangle)$$

   Let us perform measurement on the second bit now, it collapses into $|0\rangle$ with probability $\frac{1}{2}$ and $|1\rangle$ with probability $\frac{1}{2}$. After measuring the second bit let us measure the first bit.

$$P(0) = P(2^{nd}bit = |0\rangle)|\langle 0|U0\rangle|^2 + P(2^{nd}bit = |1\rangle)|\langle 0|U1\rangle|^2$$
$$P(0) = \frac{1}{2}|\langle 0|U0\rangle|^2 + \frac{1}{2}|\langle 0|U1\rangle|^2 = \frac{1}{2}|\langle U^\dagger 0|0\rangle|^2 + \frac{1}{2}|\langle U^\dagger 0|1\rangle|^2$$
$$P(1) = \frac{1}{2}|\langle 1|U0\rangle|^2 + \frac{1}{2}|\langle 1|U1\rangle|^2 = \frac{1}{2}|\langle U^\dagger 1|0\rangle|^2 + \frac{1}{2}|\langle U^\dagger 1|1\rangle|^2$$

   This can be extended to multiple coin gates.

   Hence, the probability distributions are same. The circuit indeed produces a random bit with each $\frac{1}{2}$ probability

Transformation from C to C' is given as follows:

- r COIN gates → r H + r CNOT gates + 2r ancilla bits + r output bits
- Hence C' has n input bits, a+2r anicilla bits, s+2r gates (CCNOT, CNOT and H) and m+r output bits.

Problem 3: Classical Comparison (3 points)

Give a randomized classical algorithm that makes only two queries to f, and decides the Deutsch-Jozsa problem with success probability at least 2/3 on every possible input.

Solution: Let's assume that occurrence of a constant and balanced function are equally likely. Let us query the oracle function O_f twice to get the outputs u and v respectively.

We define a classical random algorithm that decides the Deutsch Josza problem as follows:

1. if $u \neq v$, we declare that the function is balanced
2. if $u = v$, we roll a 3 sided die 3 times,we declare that the function is balanced if the outcome is side A or B and constant otherwise

Let us now analyse the probability that the declaration is correct:

- $$P(\frac{success}{Balanced}) = P(\frac{u=v}{balanced}) * P(\frac{success}{u=v}) + P(\frac{u \neq v}{balanced}) * P(\frac{success}{u \neq v})$$
$$P(\frac{success}{Balanced}) = \frac{1}{2} * P(\frac{success}{u=v}) + \frac{1}{2} * P(\frac{success}{u \neq v})$$
$$P(\frac{success}{Balanced}) = \frac{1}{2} * \frac{1}{3} + \frac{1}{2} * 1 = \frac{2}{3}$$
- $$P(\frac{success}{Constant}) = P(\frac{u=v}{constant}) * P(\frac{success}{u=v}) + P(\frac{u \neq v}{constant}) * P(\frac{success}{u \neq v})$$
$$P(\frac{success}{constant}) = 0 * P(\frac{success}{u=v}) + 1 * P(\frac{success}{u \neq v})$$
$$P(\frac{success}{constant}) = 1 * \frac{2}{3} = \frac{2}{3}$$

Note:

- When 3 sided die is rolled, P(A or B)= 2/3
- When 3 sided die is rolled, P(C)= 1/3

Hence, the proposed solution takes only 2 queries and decides the Deutsch's Josza problem with a success probability of 2/3.

Problem 4: Fun With Deutsch Josza (4 points)

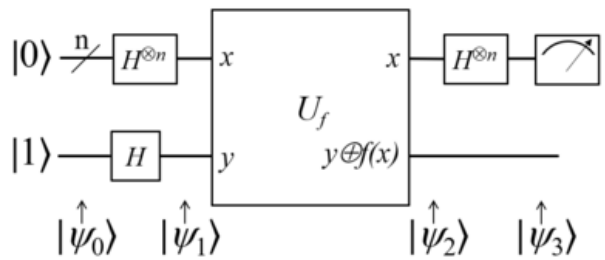
Let $N = 2^n$. Suppose our function $f : \{0,1\}^n \to \{0,1\}$ satisfies the following premise: either

1. the first $\frac{N}{2}$ bits of the truth table of f are all 0 and the second $\frac{N}{2}$ bits are all 1; or
2. the number of 1s in the first half of the truth table of f plus the number of 0s in the second half equals $\frac{N}{2}$.

Modify the Deutsch-Jozsa algorithm to efficiently distinguish between these two cases (1) and (2).

Solution:

We keep the Deutsch's Josza circuit unchanged in this proposed solution:



Let's analyse the states marked above:

$$\psi_0 = |0^n\rangle\,|1\rangle$$

We know that $H^{\otimes n}$ applied to any input $|x\rangle$ gives a superposition state given by

$$H^{\otimes n}\,|x\rangle = \frac{1}{2^{n/2}} \sum_{z\in\{0,1\}^n} (-1)^{x.z}\,|z\rangle$$

Therefore,

$$\psi_1 = \frac{1}{2^{n/2}} \sum_{z\in\{0,1\}^n} |z\rangle\,|-\rangle$$

After the oracle function f, we have

$$\psi_2 = \frac{1}{2^{n/2}} \sum_{z\in\{0,1\}^n} (-1)^{f(z)}\,|z\rangle\,|-\rangle$$

After Hadamard H on first n bits we have,

$$\psi_3 = \frac{1}{2^n} \sum_{z\in\{0,1\}^n} \sum_{x\in\{0,1\}^n} (-1)^{f(z)+x.z}\,|x\rangle\,|-\rangle$$

Drawing hints from the question, the function sets a trade off between the first N/2 bits and the next N/2 bits. Therefore, we choose to analyse the state $|100000.....0\rangle$ which is nearly halfway in the N states.

Clearly, x.z= 0 for the first N/2 bits and x.z=1 for the second N/2 bits.

1. Let the first $\frac{N}{2}$ bits of the truth table of f are all 0 and the second $\frac{N}{2}$ bits are all 1. After measurement of first n bits for $|100000.....0\rangle$, the state collapses to

$$\psi_4 = \frac{1}{2^n} \sum_{z\in\{0,1\}^n} (-1)^{f(z)+x.z}\,|100....0\rangle\,|-\rangle$$

For the first N/2 bits f(z)+x.z=0+0=0,
and the second N/2 bits f(z)+x.z= 1+1=2
Therefore, measurement gives 1 amplitude as shown below.

$$\psi_4 = \frac{1}{2^n}(2^{n-1})((-1)^0 + (-1)^2)\,|100....0\rangle\,|-\rangle = 1.\,|100...0\rangle\,|-\rangle$$

2. Let the number of 1s in the first half of the truth table of f plus the number of 0s in the second half equals $\frac{N}{2}$. After measurement of first n bits for $|100000.....0\rangle$, the state collapses to

$$\psi_4 = \frac{1}{2^n}\sum_{z\in\{0,1\}^n}(-1)^{f(z)+x.z}\,|100....0\rangle\,|-\rangle$$

Let k be the number of 1s in first half, then N/2-k is the number of 0s in the first half. We can verify that the number of 1s and 0s in second half is equal to k and N/2-k respectively.

$$\psi_4 = \frac{1}{2^n}(k*(-1)^{1+0}+(N/2-k)*(-1)^{0+0}+k*(-1)^{1+1}+(N/2-k)*(-1)^{0+1})\,|100....0\rangle\,|-\rangle$$

Therefore, measurement gives 0 amplitude as shown below.

$$\psi_4 = 0.\,|100....0\rangle\,|-\rangle$$

Therefore by measuring the state $|100....0\rangle$ in the output and getting an

- amplitude 1 indicates that the first $\frac{N}{2}$ bits of the truth table of f are all 0 and the second $\frac{N}{2}$ bits are all 1
- amplitude 0 indicates that the number of 1s in the first half of the truth table of f plus the number of 0s in the second half equals $\frac{N}{2}$

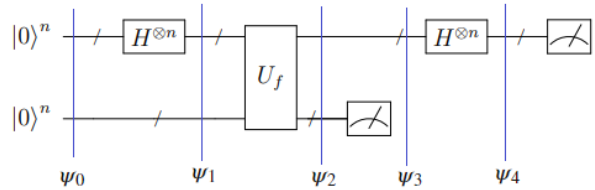
Problem 5: Simon's Algorithm

Suppose we run Simon's algorithm for the following function

$$f : \{0,1\}^3 \to \{0,1\}^3 :$$

$$f(000) = f(111) = 000,$$
$$f(001) = f(110) = 001,$$
$$f(010) = f(101) = 010,$$
$$f(011) = f(100) = 011.$$

Note that f is 2-to-1 and $f(x) = f(x \oplus 111)$ for all $x \in \{0,1\}^3$, so $s = 111$.



(a) (1 point) Give the starting state of Simon's algorithm.

Solution: $|\psi_0\rangle = |000000\rangle$

(b) (1 point) Give that state after the first Hadamard transforms on the first 3 qubits.

Solution:

$$H^{\otimes n}|x\rangle = \frac{1}{2^{n/2}} \sum_{z\in\{0,1\}^n} (-1)^{x.z}|z\rangle$$

Applying this to first 3 bits of $|000000\rangle$, we get

$$|\psi_1\rangle = \frac{1}{2^{3/2}} \sum_{z\in\{0,1\}^3} |z\rangle \otimes |000\rangle$$

(c) (2 points) Give the state after applying the oracle.

Solution:

After applying the orcale the last 3 bits correspond to f(z), hence we get

$$|\psi_2\rangle = \frac{1}{2^{3/2}} \sum_{z\in\{0,1\}^3} |z\rangle \otimes |f(z)\rangle$$

(d) (1 points) Give the state after measuring the second register (suppose the mea- surement gave $|001\rangle$.

Solution: It's clear form the function definition in question that only 2 states give output $|001\rangle$ [f (001) = f (110) = 001]

Therefore measurement giving $|001\rangle$ results a collapsed state given by,

$$|\psi_3\rangle = \frac{1}{\sqrt{2}}(|001\rangle + |110\rangle) \otimes |001\rangle$$

(e) (2 points) Using

$$H^{\otimes n}|x\rangle = \frac{1}{2^{n/2}} \sum_{z\in\{0,1\}^n} (-1)^{x.z}|z\rangle$$

, give the state after the final Hadamards.

Solution:

$$|\psi_4\rangle = \frac{1}{\sqrt{2}}(H^{\otimes n}|001\rangle + H^{\otimes n}|110\rangle) \otimes |001\rangle$$

On expansion using the formula given we get,

$$|\psi_4\rangle = \frac{1}{2^2}(|000\rangle - \cancel{|001\rangle} + \cancel{|010\rangle} - |011\rangle + \cancel{|100\rangle} - |101\rangle + |110\rangle - |111\rangle + |000\rangle + \cancel{|001\rangle} -$$

$$\cancel{|010\rangle} - |011\rangle - \cancel{|100\rangle} - |101\rangle + |110\rangle + \cancel{|111\rangle}) \otimes |001\rangle$$

$$|\psi_4\rangle = \frac{1}{2^2}(2|000\rangle - 2|011\rangle - 2|101\rangle + 2|110\rangle) \otimes |001\rangle$$

$$|\psi_4\rangle = \frac{1}{2}(|000\rangle - |011\rangle - |101\rangle + |110\rangle) \otimes |001\rangle$$

(f) (1 points) Why does a measurement of the first 3 qubits of the final stage give information about s?

Solution: Measurement of first 3 qubits gives a non-zero amplitude for all the 3 bit qubit states that are orthogonal to s.

(g) (2 points) Suppose the first run of the algorithm gives z = 011 and a second run gives z = 101. Show that, assuming s $\neq$ 000, those two runs of the algorithm already determine s.

Solution:

Let s=$a_1a_2a_3$

The z states obtained are orthogonal to s, i.e, z.s=0(mod 2)

Now, z=011, gives the condition, $a_2 + a_3 = 0 mod 2$ This means that either $a_2 = a_3 = 0$ or $a_2 = a_3 = 1$, since $s \neq 000$, we have $a_2 = 1 \& a_3 = 1$

If we get z=101 in the next run, we have $a_1 + a_3 = 0 mod 2$ This means that either $a_1 = a_3 = 0$ or $a_1 = a_3 = 1$, since $s \neq 000$, we have $a_1 = 1 \& a_3 = 1$

Constraining on both the results we have,$a_1 = 1, a_2 = 1, a_3 = 1$. Hence, s=111.

Therefore, algorithm already determines s.

Problem 6: Amplifying Success (4 points)

Let $f : \{0,1\}^n \to \{0,1\}$ be a function and C be a randomized (or quantum) circuit that computes the function f in the sense that for every input x, it holds that

$$\Pr(C(x) = f(x)) \geq \frac{2}{3}.$$

In class, we discussed how by repeating this computation several times, we can make the probability of successful computation arbitrarily large. Formalize this. In more detail, design a circuit C' such that

$$\Pr(C'(x) = f(x)) \geq 1 - 2^{-n}.$$

Solution:

Let's repeat the computation k (k is odd by assumption) times say, and let us define that the **majority occurrence is the output**.

Therefore, we can write

$$\Pr(C'(x) \neq f(x)) = \sum_{i=0}^{[\frac{k}{2}]} C_i \frac{2}{3}^i \frac{2}{3}^{k-i} = \frac{1}{3^k} \sum_{i=0}^{[\frac{k}{2}]} C_i 2^i$$

We know that $\sum_{i=0}^{k} C_i = 2^k$, based on the symmetry in binomial coefficients we can write,

$$\sum_{i=0}^{[\frac{k}{2}]} C_i = 2^{k-1}$$

Using this we can write,

$$2^{k-1} \leq \sum_{i=[\frac{k}{2}]+1}^{k} C_i 2^i \leq 2^{k+[\frac{k}{2}]}$$

Hence we can write ,

$$\frac{2^{k-1}}{3^k} \leq \Pr(C'(x) \neq f(x)) \leq \frac{2^{k-1+[\frac{k}{2}]}}{3^k}$$

This gives us,

$$1 - \frac{2^{k-1+[\frac{k}{2}]}}{3^k} \leq \Pr(C'(x) = f(x)) \leq 1 - \frac{2^{k-1}}{3^k}$$

Hence,

$$\Pr(C'(x) = f(x)) \geq 1 - \frac{2^{k-1+[\frac{k}{2}]}}{3^k}$$

To make ,

$$\Pr(C'(x) = f(x)) \geq 1 - 2^{-n}$$

we need to make,

$$1 - \frac{2^{k-1+[\frac{k}{2}]}}{3^k} \geq 1 - 2^{-n}$$

On computation and substituting $[\frac{k}{2}] = \frac{k-1}{2}$ (since k is odd) we get,

$$k \geq \frac{1}{log_2 \frac{3}{2\sqrt{2}}}(n - \frac{3}{2})$$

$$k \geq 3.06(n - \frac{3}{2})$$

So, let us say k = 4n+1 (odd number). Therefore, k is of the order O(n).

We can produce the same result using O(n) operations using Chernoff bound as well.

Problem 7: Leveraging Parity (3 points)

Let $N = 2^n$. A parity query to input $x \in \{0,1\}^N$ corresponds to the $(N+1)$-qubit unitary map Q_x:

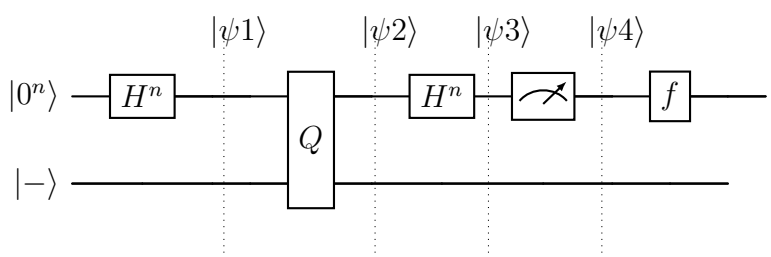
$$|y, b\rangle \to |y, b \oplus (x \cdot y)\rangle$$

where $x \cdot y = \sum_{i=0}^{N-1} x_i y_i \mod 2$.

For a fixed function $f : \{0,1\}^N \to \{0,1\}$, give a quantum algorithm that computes $f(x)$ using only one such query (i.e., one application of Q_x), and as many elementary gates as you want.

Solution:

Consider the circuit as illustrated below:



Applying H on $|0^n\rangle$ produces a superposition of all states $\in \{0,1\}^n$

$$H^{\otimes n} |0^n\rangle = \frac{1}{2^{n/2}} \sum_{z \in \{0,1\}^n} |z\rangle$$

Therefore the combined state is given by,

$$|\psi_1\rangle = \frac{1}{2^{n/2}} \sum_{z \in \{0,1\}^n} |z\rangle \otimes |-\rangle$$

After applying the oracle Q_x, we have a phase kick back given by,

$$|\psi_2\rangle = \frac{1}{2^{n/2}} \sum_{z \in \{0,1\}^n} (-1)^{x.z} |z\rangle \otimes |-\rangle$$

Now we apply hadamard again on the first n-bits,

$$|\psi_3\rangle = \frac{1}{2^{n/2}} \sum_{z \in \{0,1\}^n} (-1)^{x.z} H^{\otimes n} |z\rangle \otimes |-\rangle$$

This simplifies into a double summation given by,

$$|\psi_4\rangle = \frac{1}{2^n} \sum_{z\in\{0,1\}^n} \sum_{u\in\{0,1\}^n} (-1)^{x.z}(-1)^{u.z} |u\rangle \otimes |-\rangle = \frac{1}{2^n} \sum_{z\in\{0,1\}^n} \sum_{u\in\{0,1\}^n} (-1)^{(x\oplus u).z} |u\rangle \otimes |-\rangle$$

Now let's apply measurement, and get a state say $|u_0\rangle$,

- if $x \neq u_0$, then $x \oplus u_0 \neq |0^n\rangle$, hence when we sum over all z $\in \{0,1\}^n$, we will have equal number of orthogonal and non orthogonal elements. When the dot product is raised to the power -1, the summation gives us 0.
- if $x = u_0$, then $x \oplus u_0 = |0^n\rangle$, this results in,

$$(-1)^{(x\oplus u).z} = 1 \forall z \in \{0,1\}^n$$

Hence,

$$amplitude = \frac{1}{2^n} \sum_{z\in\{0,1\}^n} \sum_{u\in\{0,1\}^n} (-1)^{(x\oplus u).z} = 1$$

Therefore, on measurement only one state comes out with probability=1, this gives us the state x.

Noe using x we can find f(x) by parsing it into oracle function f.

Problem 8: Grover's search with multiple satisfying inputs

In this problem, we will generalize Grover's search to the case when f has more than one satisfying input. So suppose we are given an oracle O_f for $f : \{0,1\}^n \to \{0,1\}$. Define $A = \{x \mid f(x) = 1\}$ and $k = |A|$. We will assume $k \geq 1$. Also, define $B = \{x \mid f(x) = 0\}$, so $|B| = N - k$ where $N = 2^n$.

(a) (2 points) Explain how we can use the oracle to check if a given string y $\in \{0,1\}^n$ has f(y) = 1. Explain how we can use this to find an x $\in$ A with high probability in O(1) queries whenever $k >= N/2$. (We henceforth assume $k < N/2$.)

Solution: We can parse y into the oracle function to find if f(y)=1 or 0.
If we randomly choose a string $x \in \{0,1\}^n$

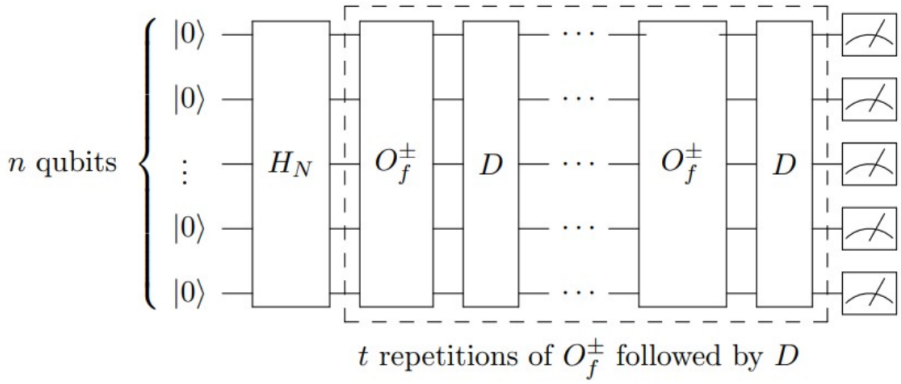
P(f(x)=1) = $\frac{k}{N}$

Since, $k >= \frac{N}{2}$, P(f(x)=1) $>= \frac{1}{2}$

(b) (2 points) Recall Grover's algorithm, with t repetitions: Letting $|\psi(t)\rangle$ denote the state of the circuit after t repetitions, we can write it as

$$|\psi(t)\rangle = \alpha_t \frac{1}{\sqrt{k}} \sum_{x\in A} |x\rangle + \beta_t \frac{1}{\sqrt{N-k}} \sum_{x\in B} |x\rangle,$$

where α_t and β_t are real numbers satisfying $\alpha_t^2 + \beta_t^2 = 1$.



What are α_0 and β_0?

Solution: The initial state is given by,

$$|\psi(t)\rangle = H^{\otimes n}\,|0^n\rangle = \frac{1}{2^{n/2}} \sum_{Z\in\{0,1\}^n} |z\rangle$$

Since, $N = 2^n$,

$$|\psi(t)\rangle = \frac{1}{\sqrt{N}} \sum_{Z\in\{0,1\}^n} |z\rangle = \frac{1}{\sqrt{N}} \sum_{x\in A} |x\rangle + \frac{1}{\sqrt{N}} \sum_{x\in B} |x\rangle$$

Comparing it with the expression of $|\psi(t)\rangle$, we get,

$$\alpha_0 \frac{1}{\sqrt{k}} = \frac{1}{\sqrt{N}}$$

$$\beta_0 \frac{1}{\sqrt{N-k}} = \frac{1}{\sqrt{N}}$$

This gives us, $\alpha_0 = \sqrt{\frac{k}{N}}$ and $\beta_0 = \sqrt{\frac{N-k}{N}}$

(c) (2 points) Thinking of (β_t, α_t) as a point on the unit circle in R2, let us write θ_t for its angle from the horizontal axis so that $(\alpha_t = \sin\theta_t, \beta_t = \cos\theta_t)$. Show that the transformation $(\beta_t, \alpha_t) - - > (\beta_{t+1}, \alpha_{t+1})$ is precisely rotation around the circle by an angle of $2\theta_0$

Solution: We can find the mean μ_t after applying the oracle function as ,

$$\mu_t = \frac{N-k}{N} \frac{\beta_t}{\sqrt{N-k}} - \frac{k}{N} \frac{\alpha_t}{\sqrt{k}}$$

$$\mu_t = \frac{\sqrt{N-k}}{N} \beta_t - \frac{\sqrt{k}}{N} \alpha_t$$

After the reflection about mean we get,

$$\beta_{t+1} = \sqrt{N-k}(2\mu_t - \frac{\beta_t}{\sqrt{N-k}}) = 2\mu_t\sqrt{N-k} - \beta_t$$

Substituting the value of μ_t,

$$\beta_{t+1} = 2\frac{N-k}{N}\beta_t - 2\frac{\sqrt{k(N-k)}}{N}\alpha_t - \beta_t$$

$$\beta_{t+1} = (\frac{2(N-k)}{N} - 1)\beta_t - 2\frac{\sqrt{k(N-k)}}{N}\alpha_t$$

Similarly, after reflection about mean for α_{t+1} we get,

$$\alpha_{t+1} = \sqrt{k}(2\mu_t + \frac{\beta_t}{\sqrt{k}}) = 2\mu_t\sqrt{k} + \alpha_t$$

Substituting the value of μ_t,

$$\alpha_{t+1} = 2(\frac{\sqrt{k(N-k)}}{N}\beta_t - \frac{k}{N}\alpha_t) + \alpha_t = \frac{2\sqrt{k(N-k)}}{N}\beta_t + (1 - 2\frac{k}{N})\alpha_t$$

Hence, we can write the transformation matrix as ,

$$\begin{bmatrix}\beta_{t+1}\\ \alpha_{t+1}\end{bmatrix} = \begin{bmatrix}\frac{2(N-k)}{N} - 1 & -2\frac{\sqrt{k(N-k)}}{N}\\ \frac{2\sqrt{k(N-k)}}{N} & 1 - 2\frac{k}{N}\end{bmatrix}\begin{bmatrix}\beta_t\\ \alpha_t\end{bmatrix}$$

By using the previous subdivision we know that

$\sin\theta_0 = \sqrt{\frac{k}{N}}$ and $\cos\theta_0 = \sqrt{\frac{N-k}{N}}$, we can write

$\frac{2(N-k)}{N} - 1 = 2\cos^2\theta_0 - 1 = \cos 2\theta_0$

$2\frac{\sqrt{k(N-k)}}{N} = 2\sin\theta_0\cos\theta_0 = \sin 2\theta_0$

$1 - 2\frac{k}{N} = 1 - 2\sin^2\theta_0 = \cos 2\theta_0$

We can write the rotation matrix as ,

$$\begin{bmatrix}\beta_{t+1}\\ \alpha_{t+1}\end{bmatrix} = \begin{bmatrix}\cos 2\theta_0 & -\sin 2\theta_0\\ \sin 2\theta_0 & \cos 2\theta_0\end{bmatrix}\begin{bmatrix}\beta_t\\ \alpha_t\end{bmatrix}$$

(d) (2 points) Assume that the algorithm knows k. Briefly, why would the algorithm like to choose

$$t = \frac{1}{2}(\frac{\pi}{2\theta} - 1)$$

if it could? Show that if it takes t to be the closest integer to this value, the circuit has the property that it outputs an element of A with a probability of at least 1/2.

Solution: If we assume it to be a unit circle, then the moving point should get closer to the y-axis, This means that $\theta_t = \frac{\pi}{2}$.

By using the rotation property, we have $\theta_t = \theta_{t-1} + 2\theta_0$ which is an Arithmetic progression .

$$\theta_t = \theta_0 + 2t\theta_0$$

Equating θ_t to $\pi/2$ we have

$$t = \frac{1}{2}(\frac{\pi}{2\theta_0} - 1)$$

If $k = N/2$, then $\theta_0 = \pi/4$, then $t = 1/2$. Taking t =1 or 2 will never lead us to the y axis. Hence it will never amplify the elements in set A. If $k \neq N/2$ then it will always reach closer to the y axis and hence the probability will be at least 1/2.

(e) (2 points) Again, assuming the algorithm knows k, show that it can find an element of A with high probability using O($\sqrt{\frac{N}{k}}$) queries to the oracle. (You may want to use that $sin\theta <= \theta$ for all $\theta >= 0$.)

Solution:

We know that,

$$sin\theta_0 \leq \theta_0$$

Subsitituting, $\sin\theta_0 = \sqrt{\frac{k}{N}}$, we get

$$\sqrt{\frac{k}{N}} \leq \theta_0$$

Now, $t = \frac{1}{2}(\frac{\pi}{2\theta_0} - 1)$

$$\sqrt{\frac{N}{k}} \geq \frac{1}{\theta_0}$$

This gives us,

$$t = \frac{1}{2}(\frac{\pi}{2\theta_0} - 1) \geq \frac{\pi}{4}\sqrt{\frac{N}{K}} - \frac{1}{2}$$

Therefore t is of the order $O(\sqrt{\frac{N}{k}})$

Problem 9: Application of Grover's algorithm (4 points)

Let N = 2^n and $x_0, , x_{N1}$ be a sequence of distinct integers (you can think of them as the outputs in the truth table of some function F). We can query this function in the usual way, i.e., we can

apply unitary O: $|0, i\rangle -> s\,|i, x_i\rangle$, as well as its inverse. The minimum of F is define is min $\{x_i | i \in \{0,\cdots,N1\}\}$. Give a quantum algorithm that finds (with probability $\geq 2/3$) an index achieving the minimum, using O($\sqrt{N}$ log N) queries. Hint: start with m = xi for a random i, and repeatedly use Grover's algorithm to find an index j such that $x_j < m$ and update $m = x_j$. Continue this until you can find no element smaller than m, and analyze the number of queries of this algorithm. You are allowed to argue about this algorithm on a high level. Bonus: give a quantum algorithm that uses O($\sqrt{N}$) queries.

Solution:

Let's define the Grover oracle to mark elements with $x_j < m$.

To get an search algorithm with O($\sqrt{N}$ log N) terms, we first begin with dividing the array of numbers into two halves.

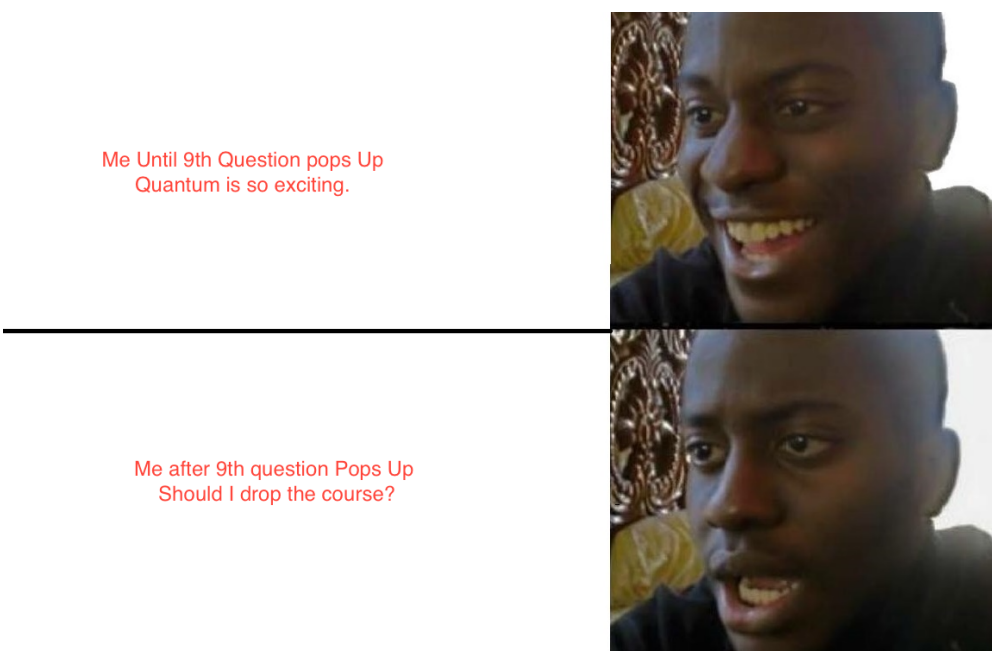
- Now if we choose a random m = xi for each half ($\frac{N}{2}$ elements each), we can find a $x_j < m$ assuming that roughly half the elements are marked up with a complexity of O($\sqrt{\frac{N}{N/2}}$)

- We repeat the experiment by assuming that the mean of the marked elements is roughly the new m. Again the grover's algorithm marks half of the previously marked elements with a complexity of O($\sqrt{\frac{N}{N/4}}$)

- We keep repeating this process (taking m to be the new mean and marking 1/2 the elements previously marked) until we reach the minimum.

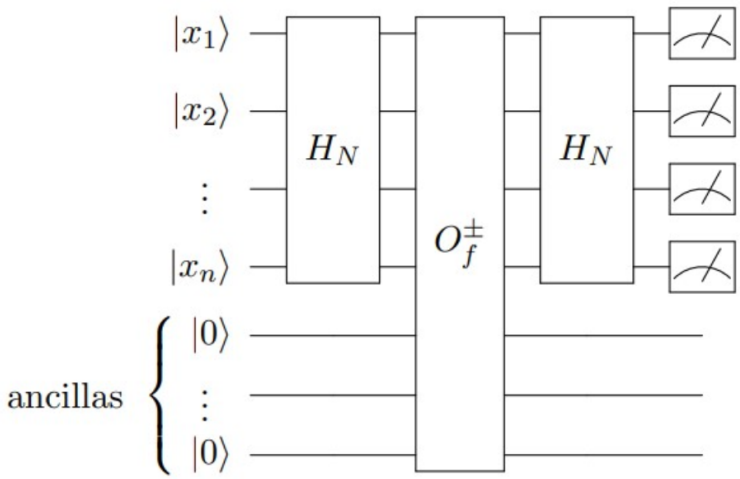
- The net complexity can be expressed as

$$O(\sqrt{2}) + O(\sqrt{4}) + O(\sqrt{8}) + O(\sqrt{16}) + ... + O(\sqrt{N}) = O(\sqrt{N})$$



Problem 10: Necessity of Uncomputing (4 points)

Recall the convention that the oracle gate $O_f^{\pm}$ for a Boolean function $f : \{0,1\}^n \to \{0,1\}$ denotes the unitary transformation $|x\rangle \mapsto (-1)^{f(x)}|x\rangle$. When implementing $O_f^{\pm}$ in some applications, we might need additional ancilla/garbage bits, in which case $O_f^{\pm}$ actually denotes the unitary transformation $|x\rangle|0^m\rangle \mapsto (-1)^{f(x)}|x\rangle|g(x)\rangle$, where $g(x)$ is whichever m-bit garbage string produced on input x. In class, we have insisted that all oracle circuits uncompute their garbage, meaning that $g(x) = 0^m$ for all $x \in \{0,1\}^n$.



In this problem, we will justify why it is okay to "pretend" the ancilla bits don't exist. We will use as our example the Deutsch–Jozsa circuit, including ancilla/garbage bits; it is drawn as follows.

Suppose $O_f^{\pm}$ uncomputes its garbage, i.e. $g(x) = 0^m$ for all x. For a given Boolean function $f : \{0,1\}^n \to \{0,1\}$ (which need not be all-0's or balanced), compute the state of the system after the $O_f^{\pm}$ gate and after the second H_n gate. Show that each of these states can be written as $|\psi_1\rangle \otimes |\psi_2\rangle$, where $|\psi_1\rangle$ is the state of the first n qubits and $|\psi_2\rangle$ is the state of the last m qubits. Finally, show that the distribution on measurement outcomes is the same as it would have been if $O_f^{\pm}$ had no ancilla bits.

Solution:

The starting state is given by,

$$|\psi_1\rangle = H^N |x\rangle \otimes |o^m\rangle = \frac{1}{2^{n/2}} \sum_{z\in\{0,1\}^n} (-1)^{x.z} |z\rangle |0^m\rangle$$

After the oracle function,

$$|\psi_2\rangle = \frac{1}{2^{n/2}} \sum_{z\in\{0,1\}^n} (-1)^{x.z+f(z)} |z\rangle |g(z)\rangle$$

Since $O_f^{\pm}$ uncomputes the garbage,

$$|\psi_2\rangle = \frac{1}{2^{n/2}} \sum_{z\in\{0,1\}^n} (-1)^{x.z+f(z)} |z\rangle |0^m\rangle$$

Since uncomputation results in the same state,

$$|\psi_2\rangle = (\frac{1}{2^{n/2}} \sum_{z\in\{0,1\}^n} (-1)^{x.z+f(z)} |z\rangle) \otimes |0^m\rangle = |\phi_1\rangle \otimes |\phi_2\rangle$$

After applying hadamard of $|\psi_2\rangle$ we get,

$$|\psi_3\rangle = (\frac{1}{2^n} \sum_{z\in\{0,1\}^n} \sum_{u\in\{0,1\}^n} (-1)^{x.z+f(z)+u.z} |u\rangle) \otimes |0^m\rangle = |\phi_3\rangle \otimes |\phi_4\rangle$$

Therefore, both the states are un-entangled.

$$|\psi_3\rangle = (\frac{1}{2^n} \sum_{z\in\{0,1\}^n} \sum_{u\in\{0,1\}^n} (-1)^{(x\oplus u).z+f(z)} |u\rangle) \otimes |0^m\rangle$$

When the first n bits are measured, and the first n bits collapse to a state say $|u_0\rangle$,

$$amplitude = (\frac{1}{2^n} \sum_{z\in\{0,1\}^n} (-1)^{(x\oplus u_0).z+f(z)})$$

$$probability = |\langle u_0|\psi_3\rangle|^2 = |\frac{1}{2^n} \sum_{z\in\{0,1\}^n} (-1)^{(x\oplus u_0).z+f(z)}|^2$$

Therefore, the probabaility distribution is independent of the garbage bits due to uncomputing.